



Project Name:

SCAFFOLD DESIGN REPORT

Scaffold Type:

Independent Return

Calculation Sheet

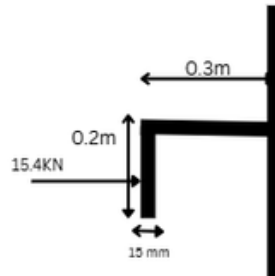
Document Number:

PMS-WA-ARCO-050

Prepared By:

NNAMANI,  
IFEANYI

## LOCAL CHECK



### 1. GIVEN

Horizontal force,  $F_X$  = 5.654 kN  
Horizontal force,  $F_Z$  = 14.276 kN  
Horizontal projection from tank,  $d$  = 0.200 m  
Vertical stiffener depth = 0.200 m  
Plate thickness = 15 mm (0.015 m)  
Stiffener material (yield strength) = 235 N/mm<sup>2</sup>

### 2. RESULTANT HORIZONTAL FORCE

$$F = \sqrt{(F_X^2 + F_Z^2)} = \sqrt{(5.654^2 + 14.276^2)}$$

$$F = 15.35 \text{ kN}$$

### 3. MOMENT AT TANK CONNECTION

$$M = F \times d = 15.35 \times 0.200$$

$$M = 3.07 \text{ kNm}$$

### 4. CHECK OF 15 mm × 200 mm STIFFENER

$$\text{Section modulus, } Z = bd^2/6 = (15 \times 200^2)/6 \\ = 100,000 \text{ mm}^3$$

$$\text{Bending stress, } \sigma_b = M/Z = (3.07 \times 10^6)/100,000$$

$$\sigma_b = 30.7 \text{ N/mm}^2$$

$$\text{Allowable bending stress (S235)} = 235 \text{ N/mm}^2$$

$$\text{Utilization ratio (bending)} = 30.7 / 235 = 0.13 < 1.00$$

OK

### 5. SHEAR CHECK

$$\text{Shear area, } A_v = t \times d = 15 \times 200 = 3,000 \text{ mm}^2$$

$$\text{Average shear stress, } \tau = F/A_v = 15,350 / 3,000$$

$$\tau = 5.12 \text{ N/mm}^2$$

$$\text{Allowable shear stress (S235)} \approx 135 \text{ N/mm}^2$$

$$\text{Utilization ratio (shear)} = 5.12 / 135 = 0.04 < 1.00$$

OK

### 6. CONCLUSION

The 15 mm × 200 mm stiffener is adequate to resist the resultant horizontal force of 15.35 kN.

Bending utilization = 0.13 < 1.00 (OK)    Shear utilization = 0.04 < 1.00 (OK)



|                  |                 |                        |                   |
|------------------|-----------------|------------------------|-------------------|
| Project Name:    |                 | SCAFFOLD DESIGN REPORT |                   |
| Scaffold Type:   |                 | Independent Return     | Calculation Sheet |
| Document Number: | PMS-WA-ARCO-050 | Prepared By:           | NNAMANI, IFEANYI  |

**CONCLUSION**

The analysis of the scaffold system shows that the system is adequate in terms of structural configuration and geometry for the purpose of its usage. The stress utility ratio shows that the scaffold pipes are working at less than unity.

**RECOMMENDATIONS:**

The scaffold shall be inspected by the Scaffold Inspector after erection to confirm that the erected scaffold conforms to design specification and working drawings before usage. Scaffold erection shall comply with design intent and material specification. Any change shall require a re-design and the scaffold Engineer shall be duly informed.

**INSTALLATION NOTE**

1. The scaffold working drawings (Plan, Elevation-Developed sheets A-D, Section A-A) and the purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the circular/radial scaffold configuration matches the approved drawings and correctly encircles the tank at the design radii (inner 29.7 m, mid 31.7 m, outer 33.7 m) before proceeding above the base lift.
3. Base plates must be set out and levelled at all 416 base support positions before erection of standards begins. Confirm each base plate bears fully on a sound, level surface – do not pack or wedge in place of a properly prepared base.
4. Erection shall proceed lift-by-lift (2 m per lift) working continuously around the full circumference, closing the ring back to the starting point before moving up to the next lift. Do not erect isolated tall sections ahead of the surrounding ring – this scaffold depends on ring continuity for stability.
5. Platform boarding is NOT uniform at every level – only the lowest platform (~3.7 m) is boarded continuously around the full 360°; the upper 7 platform levels are boarded only in specific arcs per the approved drawings. Scaffolders must follow the drawings exactly at each level and must not assume a level is fully boarded just because the lift below it was – confirm against the Plan/Elevation-Developed sheets before decking each level.
6. All tie points to the tank shell must be installed strictly as shown on the drawings, at the correct level and radial position, before the tied lift is loaded. Do not omit or relocate ties without engineering approval – ring stability relies on the tie pattern shown.
7. The work vicinity at the tank base must be barricaded and appropriate signage placed at all access points before erection commences.
8. All scaffolders must wear complete PPE, including fall arrest equipment where required, in accordance with site work-at-height regulations.
9. A valid work permit must be obtained, discussed, and signed before erection, modification, or dismantling commences.
10. Scaffold erection, modification, and dismantling must be carried out in strict accordance with NASC guidance (or equivalent site standard) and site safety procedures, under the supervision of a competent Scaffold Supervisor.
11. On completion of each lift, the erected work shall be checked for plumb, level, and correct tie engagement before the next lift proceeds.
12. The scaffold shall be inspected by a competent person and tagged before first use, and re-inspected at the frequency required by site procedure or after any incident that could affect its structural integrity (e.g. adverse weather, impact).